

LifeLink: Organ Donation Management System

Project Proposal

1. PROJECT OVERVIEW

Project Title: LifeLink - Organ Donation Coordination Web Platform

Project Type: Web Application (Full Stack)

Domain: Healthcare Information Management

Academic Level: Undergraduate Final Year Project

2. INTRODUCTION

Organ donation is a critical healthcare challenge worldwide. Thousands of patients die annually waiting for organ transplants due to inefficient coordination between donors, recipients, and medical facilities. LifeLink addresses this gap by providing a centralized digital platform to streamline organ donation registration, matching, and coordination processes.

Problem Statement:

- Lack of centralized donor registration systems
- Inefficient communication between donors and hospitals
- Limited public awareness about organ donation
- Manual, time-consuming matching processes
- No transparent tracking of donation requests

Proposed Solution: A web-based management system that enables:

- Easy donor registration and consent management
 - Efficient organ request processing for hospitals
 - Automated donor-recipient matching based on criteria
 - Administrative oversight and reporting
 - Public awareness and education resources
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3. OBJECTIVES

Primary Objectives:

1. Develop a secure, role-based web application for organ donation management
2. Implement donor registration with consent tracking and profile management
3. Create a hospital interface for organ requests and donor searches
4. Build an administrative dashboard for verification and system oversight
5. Provide awareness resources to increase donation rates

Secondary Objectives:

1. Implement real-time notifications for critical requests
 2. Generate analytical reports and statistics
 3. Ensure data privacy and GDPR-compliant information handling
 4. Create responsive, accessible user interfaces
 5. Develop RESTful APIs for potential third-party integration
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4. SCOPE

In Scope:

- User authentication and authorization (JWT-based)

- Donor registration and profile management
- Organ request submission and tracking
- Admin verification and approval workflows
- Search and filter functionality
- Email notifications
- Dashboard with statistics
- Awareness content management
- Data export capabilities

Out of Scope:

- Real medical diagnosis or clinical decision-making
- Direct medical device integration
- Actual organ allocation decisions (this is a coordination tool)
- Financial transactions or payment processing
- Telemedicine or video consultation features
- Mobile native applications (responsive web only)

Important Note: This system is designed as an **information and coordination platform** to facilitate communication between stakeholders. Final medical decisions remain with qualified healthcare professionals and regulatory bodies.

5. SYSTEM USERS

1. Donor (Public User)

- Register as potential organ donor
- Create and update donor profile
- Select organs willing to donate
- Manage consent status
- View donation awareness content
- Receive notifications about system updates

2. Hospital Representative / Recipient Coordinator

- Submit organ requests for patients
- Search available donors (with anonymized data)
- Track request status
- View matched donors
- Communicate with admin for approvals

3. System Administrator

- Verify donor registrations
 - Review and approve organ requests
 - Manage donor-recipient matching
 - Generate system reports
 - Manage user accounts
 - Monitor system health and activity
 - Update awareness content
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6. FUNCTIONAL REQUIREMENTS

6.1 Authentication & Authorization

- FR-1: User registration with email verification

- FR-2: Secure login with JWT token-based authentication
- FR-3: Role-based access control (Donor, Hospital, Admin)
- FR-4: Password recovery and reset functionality
- FR-5: Session management and auto-logout

6.2 Donor Management

- FR-6: Donor profile creation with personal details
- FR-7: Medical history input (blood type, allergies, conditions)
- FR-8: Organ selection (multiple organs supported)
- FR-9: Consent management (update/revoke consent)
- FR-10: Document upload (ID proof, medical certificates)
- FR-11: Emergency contact management

6.3 Request Management

- FR-12: Hospital can submit organ requests
- FR-13: Request includes patient details (anonymized), urgency level, organ type
- FR-14: Request status tracking (Pending, Under Review, Matched, Completed, Rejected)
- FR-15: Priority flagging for urgent cases
- FR-16: Request history and audit trail

6.4 Search & Matching

- FR-17: Advanced search with filters (organ type, blood type, location, availability)
- FR-18: Anonymized donor listing for hospitals
- FR-19: Compatibility indicators (blood type matching)
- FR-20: Location-based proximity sorting
- FR-21: Availability status display

6.5 Admin Operations

- FR-22: Donor verification workflow
- FR-23: Request approval/rejection
- FR-24: Match proposal creation
- FR-25: User management (activate/deactivate accounts)
- FR-26: System configuration management

6.6 Notifications

- FR-27: Email notifications for registration confirmation
- FR-28: Alerts for request status changes
- FR-29: Admin notifications for pending approvals
- FR-30: Emergency request notifications

6.7 Reporting & Analytics

- FR-31: Dashboard with key metrics (total donors, active requests, matches)
- FR-32: Donation statistics by region, organ type
- FR-33: Trend analysis (monthly registrations, request patterns)
- FR-34: Export reports (PDF, CSV)
- FR-35: Audit logs for compliance

6.8 Awareness & Education

- FR-36: Public information pages about organ donation
- FR-37: FAQs and eligibility guidelines
- FR-38: Success stories and testimonials

- FR-39: Blog/article management (admin-created)
 - FR-40: Downloadable educational materials
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7. NON-FUNCTIONAL REQUIREMENTS

Security:

- NFR-1: All sensitive data encrypted at rest and in transit (TLS 1.3)
- NFR-2: Password hashing using BCrypt
- NFR-3: SQL injection prevention through parameterized queries
- NFR-4: XSS and CSRF protection
- NFR-5: Regular security audits and dependency updates

Performance:

- NFR-6: Page load time < 3 seconds on 4G networks
- NFR-7: API response time < 500ms for 95% of requests
- NFR-8: Support 1000+ concurrent users
- NFR-9: Database query optimization with indexing

Usability:

- NFR-10: Intuitive, accessible UI (WCAG 2.1 AA compliant)
- NFR-11: Responsive design (mobile, tablet, desktop)
- NFR-12: Multi-language support (English primary)
- NFR-13: Clear error messages and user guidance

Reliability:

- NFR-14: 99.5% uptime SLA
- NFR-15: Automated backups (daily)
- NFR-16: Disaster recovery plan
- NFR-17: Transaction rollback on failures

Maintainability:

- NFR-18: Modular, well-documented code
- NFR-19: Comprehensive API documentation (Swagger)
- NFR-20: Version control with Git
- NFR-21: Continuous integration pipeline

Scalability:

- NFR-22: Horizontal scaling capability
 - NFR-23: Database sharding support for future growth
 - NFR-24: Caching layer for frequently accessed data
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8. TECHNOLOGY STACK

Frontend:

- **Framework:** React 18.x
- **Styling:** Tailwind CSS
- **State Management:** React Context API / Redux Toolkit
- **Routing:** React Router v6
- **HTTP Client:** Axios
- **Charts:** Chart.js / Recharts

- **Forms:** Formik + Yup validation
- **Notifications:** React Toastify

Backend:

- **Framework:** Spring Boot 3.x (Java 17+)
- **API Style:** RESTful APIs
- **Security:** Spring Security + JWT
- **ORM:** Spring Data JPA (Hibernate)
- **Validation:** Hibernate Validator
- **Email:** Spring Mail with SMTP
- **Documentation:** Swagger/OpenAPI

Database:

- **Primary DB:** MySQL 8.x
- **Connection Pooling:** HikariCP
- **Migration:** Flyway

Development Tools:

- **Build Tools:** Maven (Backend), Vite (Frontend)
- **Version Control:** Git
- **API Testing:** Postman
- **Code Quality:** SonarLint
- **IDE:** IntelliJ IDEA / VS Code

Deployment:

- **Containerization:** Docker
- **Web Server:** Nginx (reverse proxy)
- **Application Server:** Embedded Tomcat
- **CI/CD:** GitHub Actions (optional)

9. SYSTEM ARCHITECTURE

Architecture Pattern: Three-Tier Architecture



Key Components:

1. **Frontend React App:** Single Page Application (SPA)
 2. **REST API Layer:** JSON-based communication
 3. **Business Logic Layer:** Services, validation, orchestration
 4. **Data Access Layer:** Repository pattern with JPA
 5. **Security Layer:** JWT authentication, role-based authorization
 6. **Email Service:** Asynchronous notification delivery
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10. DATABASE DESIGN

Core Entities:

1. **User** - Authentication and common user data
2. **Donor** - Donor-specific profile and medical info
3. **Organ** - Available organs for donation
4. **Request** - Organ requests from hospitals
5. **Match** - Proposed donor-recipient matches
6. **AuditLog** - System activity tracking

Key Relationships:

- User (1) → (0..1) Donor
- Donor (1) → (M) Organ
- Request (M) → (1) Organ Type
- Match (M) → (1) Donor
- Match (M) → (1) Request

(Detailed ER Diagram provided in separate document)

11. PROJECT DELIVERABLES

Phase 1 - Planning & Design (Weeks 1-2)

- ✓ Project proposal document
- ✓ Requirements specification
- ✓ System architecture design
- ✓ Database schema and ER diagram
- ✓ UI/UX wireframes
- ✓ API endpoint documentation

Phase 2 - Backend Development (Weeks 3-6)

- ✓ Database setup and migrations
- ✓ User authentication and authorization
- ✓ Donor management APIs
- ✓ Request management APIs
- ✓ Search and matching logic
- ✓ Admin operations APIs
- ✓ Email notification service
- ✓ Unit and integration tests

Phase 3 - Frontend Development (Weeks 7-10)

- ✓ Component library setup
- ✓ Authentication UI (Login, Register)
- ✓ Donor dashboard and forms
- ✓ Hospital request interface

- ✓ Admin panel
- ✓ Search and filter pages
- ✓ Awareness content pages
- ✓ Responsive design implementation

Phase 4 - Integration & Testing (Weeks 11-12)

- ✓ Frontend-backend integration
- ✓ End-to-end testing
- ✓ Security testing
- ✓ Performance optimization
- ✓ Bug fixes

Phase 5 - Deployment & Documentation (Weeks 13-14)

- ✓ Production deployment
- ✓ User manual
- ✓ Technical documentation
- ✓ Final project report
- ✓ Presentation slides

12. RISK ANALYSIS

Risk	Probability	Impact	Mitigation Strategy
Data privacy breach	Low	Critical	Implement encryption, regular security audits, compliance with data protection laws
Scope creep	Medium	High	Strict requirement management, change control process
Technical skill gaps	Medium	Medium	Online courses, documentation review, mentor consultation
Third-party API failures	Low	Medium	Fallback mechanisms, error handling, local caching
Performance issues	Medium	High	Load testing, optimization, caching strategies
Delayed timeline	Medium	High	Agile sprints, weekly progress reviews, buffer time

13. ETHICAL CONSIDERATIONS

1. **Data Privacy:** All personal and medical data is encrypted and access-controlled. Only authorized personnel can view sensitive information.
2. **Informed Consent:** Donors are clearly informed about how their data will be used. Consent can be updated or withdrawn at any time.
3. **No Medical Decisions:** System explicitly states it does not make clinical decisions. It is a coordination tool only.
4. **Non-Discrimination:** Search algorithms do not discriminate based on age, gender, race, or socioeconomic status beyond medically necessary criteria.
5. **Transparency:** Clear terms of service and privacy policy. Users understand system limitations.
6. **Compliance:** Adherence to relevant healthcare data regulations (HIPAA-inspired practices, GDPR for European users).

14. FUTURE ENHANCEMENTS

1. **AI-Powered Matching:** Machine learning algorithms for optimal donor-recipient matching

- 2. **Mobile Applications:** Native iOS and Android apps
 - 3. **Real-Time Chat:** Communication channel between hospitals and admin
 - 4. **Blockchain Integration:** Immutable audit trail for transparency
 - 5. **Multilingual Support:** Support for 10+ languages
 - 6. **Telemedicine Integration:** Video consultation capabilities
 - 7. **IoT Integration:** Medical device data integration
 - 8. **Geographical Expansion:** Multi-country support with localization
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15. SUCCESS CRITERIA

Technical Metrics:

- All functional requirements implemented (100%)
- Zero critical security vulnerabilities
- <500ms average API response time
- 95%+ test coverage

Academic Metrics:

- Complete documentation delivered
- Successful project demonstration
- Positive evaluation from review panel

User Metrics:

- Intuitive UI with <5 min learning curve
 - Positive user feedback from testing
 - Successful donor registration flow completion >90%
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16. CONCLUSION

LifeLink represents a comprehensive solution to modernize organ donation coordination. By providing a secure, efficient platform for donors, hospitals, and administrators, this system has the potential to save lives through improved communication and streamlined processes.

The project demonstrates proficiency in full-stack web development, database design, security implementation, and user-centered design principles. It addresses a real-world problem with social impact while maintaining academic rigor and technical excellence.

Key Strengths:

-  Clear social impact and purpose
-  Well-defined technical scope
-  Modern, industry-standard tech stack
-  Comprehensive feature set
-  Scalable architecture
-  Strong emphasis on security and ethics

This project is feasible within the academic timeline while offering opportunities for impressive technical demonstrations and meaningful real-world application.

Prepared By: [Your Name]

Date: February 2026

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